

Translated

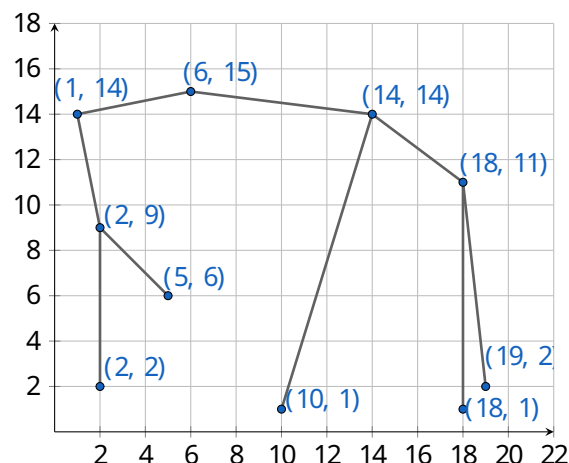
From: e-maxx.ru

Treap (Cartesian tree)

A treap is a data structure which combines binary tree and binary heap (hence the name: tree + heap $\rightarrow$ Treap).

More specifically, treap is a data structure that stores pairs (X, Y) in a binary tree in such a way that it is a binary search tree by X and a binary heap by Y . If some node of the tree contains values (X_0, Y_0) , all nodes in the left subtree have $X \leq X_0$, all nodes in the right subtree have $X_0 \leq X$, and all nodes in both left and right subtrees have $Y \leq Y_0$.

A treap is also often referred to as a "cartesian tree", as it is easy to embed it in a Cartesian plane:



Treaps have been proposed by Raimund Siedel and Cecilia Aragon in 1989.

Advantages of such data organisation

In such implementation, X values are the keys (and at same time the values stored in the treap), and Y values are called **priorities**. Without priorities, the treap would be a regular binary search tree by X , and one set of X values could correspond to a lot of different trees, some of them degenerate (for example, in the form of a linked list), and therefore extremely slow (the main operations would have $O(N)$ complexity).

At the same time, **priorities** (when they're unique) allow to **uniquely** specify the tree that will be constructed (of course, it does not depend on the order in which values are added), which can